Online Appendix
A Hundred Years of Business Cycles and the Phillips Curve

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A Derivation of the Stylised Model

This appendix reports the derivation of the model discussed in Section 2 of the paper. Appendix A.1 defines the trends and the cyclical components and delivers the three-equation model in cyclical terms. Appendix A.2 solves it for i.i.d. disturbances, discusses determinacy, and shows that a single state – the business cycle component – drives every cyclical variable. Appendix A.3 generalises the dynamics to persistent disturbances and collects the results in the representation that Section 3 takes to the data.

A.1 The economy – trends and cycles

At the very core of the study of business cycle fluctuations there is a decomposition of output (in log levels) into a trend – potential output – and a stationary component:

$$y_t = \tau_t^y + \widehat{y}_t. \tag{A.1}$$

The trend, τ_t^y, is usually thought of as determined by technological progress and by demographic and institutional factors, which inform the long-run behaviour of GDP. A commonly adopted specification for potential output is an integrated process of order two, i.e. I(2), in which its growth rate $g_t \equiv \Delta\tau_t^y$ is a driftless random walk that captures stochastic changes in trend growth:

$$\tau_t^y = \tau_{t-1}^y + g_t, \tag{A.2}$$

$$g_t = g_{t-1} + u_t^g. \tag{A.3}$$

The fluctuation of output off its trend, $\widehat{y}_t$, is instead due to the action of different structural cyclical factors – demand, supply, monetary – pushing output off its long-run equilibrium. This measure of business cycle fluctuations, in the Frisch–Slutsky paradigm, is usually considered to be representable as a mean-zero stationary stochastic process.

The analysis of business cycles has shown that fluctuations in many macroeconomic indicators – prices, financial and labour-market variables – comove with those of output, albeit with lags and leads and to a different extent. We therefore apply the same decomposition to the

other variables of the model and let their stationary components be driven by that of output. In our model we assume that unemployment can be decomposed into a trend and a cyclical component,

$$u_t = \tau_t^u + \widehat{u}_t. \tag{A.4}$$

Trend unemployment, τ_t^u – i.e. the rate consistent with output at its potential – is called equilibrium unemployment and is thought to be due to structural and institutional factors in the labour market. We model it as a driftless random walk,

$$\tau_t^u = \tau_{t-1}^u + u_t^{\tau,u}. \tag{A.5}$$

We do not explicitly model the labour market, but we assume that the stationary component of output is reflected in the cyclical component of unemployment via Okun's law, i.e.

$$\widehat{u}_t = -\delta_u(L)\widehat{y}_t, \tag{A.6}$$

where the lag polynomial $\delta_u(L)$ has non-negative coefficients, so that a positive deviation of output from its trend is associated with unemployment below its equilibrium level.

Inflation is also characterised by a trend, i.e. the inflation rate prevailing in the absence of cyclical factors, and by a cyclical component,

$$\pi_t = \tau_t^\pi + \widehat{\pi}_t, \tag{A.7}$$

where trend inflation reflects the long-term rational expectations of agents, $\tau_t^\pi = \lim_{h\to\infty} E_t\pi_{t+h}$ (Kozicki and Tinsley, 2001; Ireland, 2007; Cogley and Sbordone, 2008; Cogley et al., 2010). It coincides with the inflation target only when the central bank is fully credible, and the two can differ substantially when credibility is incomplete. We model it as a driftless random walk,

$$\tau_t^\pi = \tau_{t-1}^\pi + u_t^{\tau,\pi}. \tag{A.8}$$

The cyclical component, which we derive below, is characterised in a New Keynesian model by the Phillips curve, which connects cyclical inflation to the stationary component of output and

can be affected by different types of supply shocks.

Finally, the nominal interest rate, set by the central bank, is characterised by a nominal trend, which can be decomposed into a real interest rate trend plus the inflation trend, and by a cyclical component,

$$i_t = \tau_t^i + \widehat{\imath}_t = \tau_t^r + \tau_t^\pi + \widehat{\imath}_t. \tag{A.9}$$

The processes for τ_t^r and for the cyclical component are discussed below.

To summarise, each variable is decomposed into a trend τ_t^x and a cyclical component, which we denote by $\widehat{x}_t$. The trends of inflation, unemployment and output growth are assumed to be driftless random walks, with trend innovations that are mean zero and unpredictable. Hence these trends are martingale processes, i.e. $E_t\tau_{t+1}^x = \tau_t^x$ for $x \in \{u, \pi\}$ and $E_t g_{t+1} = g_t$.

We now provide additional structure to the model by discussing cyclical components and structural relationships. Let us start from the Fisher equation.

Expectations and the Fisher equation. Taking conditional expectations of Eq. (A.7) and using the martingale property of τ_t^π, one obtains

$$E_t\pi_{t+1} = E_t\tau_{t+1}^\pi + E_t\widehat{\pi}_{t+1} = \tau_t^\pi + E_t\widehat{\pi}_{t+1}. \tag{A.10}$$

The ex-ante cyclical component of the real interest rate – the cyclical Fisher equation – can be obtained by combining Eqs. (A.9) and (A.10), i.e.

$$\begin{aligned} r_t - \tau_t^r &\equiv i_t - E_t\pi_{t+1} - \tau_t^r \\ &= \left(\tau_t^r + \tau_t^\pi + \widehat{\imath}_t\right) - \left(\tau_t^\pi + E_t\widehat{\pi}_{t+1}\right) - \tau_t^r = \widehat{\imath}_t - E_t\widehat{\pi}_{t+1}. \end{aligned} \tag{A.11}$$

The pricing block and the Phillips curve. We model the cyclical relationships as in a standard New Keynesian model with Calvo pricing. However, the presence of non-zero trend inflation would in general entail non-standard equations with time-varying coefficients, as discussed in Ascari (2004) and Ascari and Ropele (2009) (see Ascari and Sbordone, 2014 for a review). We abstract from these issues by assuming that prices that are not reset in each period are subject to full indexation.

As discussed in Cogley and Sbordone (2008), Calvo pricing and full indexation of the non-

reset prices deliver a Phillips curve in deviations from trend, written in terms of the output gap:

$$\pi_t - \tau_t^{\pi} = \beta E_t\big(\pi_{t+1} - \tau_{t+1}^{\pi}\big) + \kappa\big(\widehat{y}_t - \widehat{y}_t^{flex}\big) + \varepsilon_t, \tag{A.12}$$

where ε_t is a cost-push shock, reflecting fluctuations in desired mark-ups, and the output gap is the difference between output and the level that would prevail under flexible prices, both in deviation from trend. The latter is a function of the transitory component of technology $\widehat{a}_t$, the permanent component being part of potential output:

$$\widehat{y}_t^{flex} = (1+\eta)(\chi+\eta)^{-1}\,\widehat{a}_t, \tag{A.13}$$

with η the inverse Frisch elasticity and χ the coefficient of relative risk aversion. Substituting Eq. (A.13) into Eq. (A.12), and collecting the cost-push shock and the technology term into a single supply disturbance,

$$u_t^s \equiv \varepsilon_t - \kappa(1+\eta)(\chi+\eta)^{-1}\,\widehat{a}_t, \tag{A.14}$$

the Phillips curve can be written in terms of the stationary component of output alone,

$$\pi_t - \tau_t^{\pi} = \beta E_t\big(\pi_{t+1} - \tau_{t+1}^{\pi}\big) + \kappa\widehat{y}_t + u_t^s. \tag{A.15}$$

The supply disturbance thus collects the two sources of variation that move inflation at a given level of $\widehat{y}_t$.

Using Eq. (A.10), $E_t(\pi_{t+1} - \tau_{t+1}^{\pi}) = E_t\pi_{t+1} - \tau_t^{\pi}$, so that the equation simplifies to

$$\pi_t = \beta E_t\pi_{t+1} + (1-\beta)\tau_t^{\pi} + \kappa\widehat{y}_t + u_t^s, \tag{A.16}$$

where the term $(1-\beta)\tau_t^{\pi}$ is due to the assumption of indexation. Substituting $\pi_t = \tau_t^{\pi} + \widehat{\pi}_t$ and Eq. (A.10) into Eq. (A.16), the trend terms cancel out and one obtains[1]

$$\widehat{\pi}_t = \beta E_t\widehat{\pi}_{t+1} + \kappa\widehat{y}_t + u_t^s. \tag{A.17}$$

The policy rule. We assume that the central bank sets the nominal interest rate in deviation

[1]The removal of the trend terms in the Phillips curve is exact if prices are fully indexed to the trend. Absent full indexation, the substitution would leave a residual $(\beta-1)\tau_t^{\pi}$, which would vanish only for $\beta = 1$.

from an anchor given by the trend of the real rate plus expected inflation, $\tau_t^r + E_t\pi_{t+1}$, by reacting to deviations of inflation from trend inflation, and possibly in response to demand shocks (which we define below), u_t^d, and to its own monetary policy shocks:

$$i_t = \tau_t^r + E_t\pi_{t+1} + \theta_\pi\big(\pi_t - \tau_t^\pi\big) + \theta_d u_t^d + u_t^{mp}, \qquad \theta_\pi > 0. \tag{A.18}$$

The anchor $\tau_t^r + E_t\pi_{t+1}$ coincides with the nominal trend τ_t^i of Eq. (A.9) up to the cyclical component of expected inflation, $E_t\widehat{\pi}_{t+1}$. Because the rule already carries expected inflation with a unit coefficient, the Taylor principle takes the form $\theta_\pi > 0$ rather than the more familiar $\theta_\pi > 1$; we return to this in Appendix A.2. Subtracting $\tau_t^r + \tau_t^\pi$ from Eq. (A.18), and using Eq. (A.10), one gets

$$\widehat{\imath}_t = E_t\widehat{\pi}_{t+1} + \theta_\pi\widehat{\pi}_t + \theta_d u_t^d + u_t^{mp}. \tag{A.19}$$

The Euler equation and the neutral rate. A hybrid Euler equation, in log-levels, is specified as

$$y_t = \alpha y_{t-1} + (1-\alpha)E_t y_{t+1} - \sigma r_t + \zeta_t, \tag{A.20}$$

with $\alpha \in (0,1)$ the degree of backward-lookingness and ζ_t a stationary preference shifter. The weights sum to one and the linearised Euler equation is a weighted average of past and expected future activity plus an intertemporal substitution term, and the weights inherit the normalisation of the underlying difference equation.

Along the trend path output is at potential and the stationary disturbance is absent. Evaluating Eq. (A.20) at $y_t = \tau_t^y$ and $\zeta_t = 0$ defines the trend of the real rate, τ_t^r, as the rate that sustains that path,

$$\tau_t^y = \alpha\tau_{t-1}^y + (1-\alpha)E_t\tau_{t+1}^y - \sigma\tau_t^r\,, \tag{A.21}$$

which delivers

$$\begin{aligned}\tau_t^r &= \frac{1}{\sigma}(1-\alpha)E_t\Delta\tau_{t+1}^y - \frac{1}{\sigma}\alpha\Delta\tau_t^y \\ &= \frac{1}{\sigma}(1-2\alpha)g_t.\end{aligned} \tag{A.22}$$

The level of potential output cancels, so only trend growth enters the trend of the real rate.

Subtracting Eq. (A.21) from Eq. (A.20) delivers

$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma\big(r_t - r_t^n\big), \tag{A.23}$$

where the neutral rate, the real rate consistent with output at potential, is defined as

$$r_t^n \equiv \tau_t^r + u_t^d = \frac{1}{\sigma}(1-2\alpha)\, g_t + u_t^d, \tag{A.24}$$

with $u_t^d \equiv \zeta_t/\sigma$. The neutral rate consists of a permanent component, the trend of the real rate, and a transitory one, the demand disturbance. The Fisher identity (A.11) replaces $r_t - \tau_t^r$ by $\widehat{\imath}_t - E_t\widehat{\pi}_{t+1}$, delivering the IS equation

$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma\big(\widehat{\imath}_t - E_t\widehat{\pi}_{t+1} - u_t^d\big). \tag{A.25}$$

The demand shock in the IS block is therefore the transitory component of the neutral rate, which is also what makes it an object the central bank can respond to in its policy rule, Eq. (A.18).

A comment on the order of integration of output. The order of integration of potential output is consequential. Because g_t is a driftless random walk, Eq. (A.22) implies that τ_t^r is an integrated process of order one, i.e. I(1).[2] This is consistent with the unit-root component assigned to the interest rate in Section 3. Consider instead assuming that the log level of potential output were integrated of order one, i.e. $\tau_t^y = \mu + \tau_{t-1}^y + u_t^{\tau,y}$. Trend growth would then be constant up to serially uncorrelated noise, $E_t\Delta\tau_{t+1}^y = \mu$ and $\Delta\tau_t^y = \mu + u_t^{\tau,y}$, and Eq. (A.22) would reduce to

$$\sigma\tau_t^r = (1-2\alpha)\mu - \alpha u_t^{\tau,y}. \tag{A.26}$$

The trend of the real rate would then be a constant plus white noise. Such a specification has no low-frequency variation in the equilibrium real interest rate, which would be at odds with the secular movements in the neutral rate documented in the literature.

It is also worth observing that Eq. (A.22) renders τ_t^r a function of trend growth, so the model

[2]The coefficient $(1-2\alpha)$ vanishes at $\alpha = \frac{1}{2}$, so at that degree of backward-lookingness the link between the neutral rate and expected trend growth would be severed altogether.

does not deliver a neutral rate that is independent of the output trend: absent the preference shifter, the two would share the innovation u_t^g exactly. Treating τ_t^r as a separate unit-root component, orthogonal to the other trends, is therefore a restriction imposed in estimation rather than an implication of the derivation.

The three-equation model. The cyclical dynamics described by the model are summarised by the three equations derived above, i.e.

$$\widehat{\pi}_t = \beta E_t \widehat{\pi}_{t+1} + \kappa \widehat{y}_t + u_t^s, \tag{A.27}$$

$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma\big(\widehat{\imath}_t - E_t \widehat{\pi}_{t+1} - u_t^d\big), \tag{A.28}$$

$$\widehat{\imath}_t = E_t \widehat{\pi}_{t+1} + \theta_\pi \widehat{\pi}_t + \theta_d u_t^d + u_t^{mp}. \tag{A.29}$$

It is worth stressing that the trend components do not appear in any of the three equations.

A.2 Solution and determinacy

We are now in a position to derive the solution of the model and to discuss its properties.

Guess and verify. From Eq. (A.29), the real-rate gap is $\widehat{\imath}_t - E_t \widehat{\pi}_{t+1} = \theta_\pi \widehat{\pi}_t + \theta_d u_t^d + u_t^{mp}$, so that Eq. (A.28) becomes

$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma \theta_\pi \widehat{\pi}_t + \sigma(1-\theta_d) u_t^d - \sigma u_t^{mp}. \tag{A.30}$$

Let us also assume that all the disturbances are i.i.d. – we relax this assumption in Appendix A.3. The only endogenous predetermined variable is $\widehat{y}_{t-1}$, and we look for a solution of the form

$$\widehat{y}_t = \gamma \widehat{y}_{t-1} + \textstyle\sum_j b_j u_t^j, \tag{A.31}$$

$$\widehat{\pi}_t = \varphi \widehat{y}_{t-1} + \textstyle\sum_j c_j u_t^j, \tag{A.32}$$

with $j \in \{d, s, mp\}$, implying $E_t \widehat{y}_{t+1} = \gamma \widehat{y}_t$ and $E_t \widehat{\pi}_{t+1} = \varphi \widehat{y}_t$.

Substituting the guess into the Phillips curve in Eq. (A.27) gives $\widehat{\pi}_t = (\beta\varphi + \kappa)\widehat{y}_t + u_t^s$, so that, substituting Eq. (A.31) and matching the coefficients on $\widehat{y}_{t-1}$ and on each shock, one

finds the conditions

$$\varphi = (\beta\varphi + \kappa)\gamma \equiv \Phi\gamma, \tag{A.33}$$

$$c_j = (\beta\varphi + \kappa)b_j + \mathbf{1}_{\{j=s\}} \equiv \Phi b_j + \mathbf{1}_{\{j=s\}}. \tag{A.34}$$

The first condition solves as

$$\varphi = \frac{\gamma\kappa}{1-\beta\gamma}, \qquad \text{so that} \qquad \Phi \equiv \beta\varphi + \kappa = \frac{\kappa}{1-\beta\gamma}, \tag{A.35}$$

and, together with Eqs. (A.33) and (A.34), it implies that cyclical inflation is proportional to the contemporaneous $\widehat{y}_t$ up to the unpredictable supply shock,

$$\widehat{\pi}_t = \Phi\widehat{y}_t + u_t^s, \tag{A.36}$$

while the policy rate, from Eq. (A.29), is a further loading on the same state, i.e.

$$\widehat{\imath}_t = \Phi(\gamma + \theta_\pi)\widehat{y}_t + \theta_\pi u_t^s + \theta_d u_t^d + u_t^{mp}. \tag{A.37}$$

Repeating the exercise on Eq. (A.30) pins down γ and the shock loadings. Substituting $E_t\widehat{y}_{t+1} = \gamma\widehat{y}_t$ and Eq. (A.36) and collecting terms gives $D\widehat{y}_t = \alpha\widehat{y}_{t-1} + \sum_j \varsigma_j u_t^j$, with $D \equiv 1 - (1-\alpha)\gamma + \sigma\theta_\pi\Phi$, $\varsigma_d = \sigma(1-\theta_d)$, $\varsigma_s = -\sigma\theta_\pi$ and $\varsigma_{mp} = -\sigma$; matching against the guess requires $\gamma D = \alpha$ and $b_j D = \varsigma_j$. Multiplying the first by $(1-\beta\gamma)$ delivers the cubic equation

$$F(\gamma) \equiv \beta(1-\alpha)\gamma^3 - (1-\alpha+\beta)\gamma^2 + \big(1 + \alpha\beta + \sigma\kappa\theta_\pi\big)\gamma - \alpha = 0, \tag{A.38}$$

while $D = \alpha/\gamma$ gives $b_j = (\gamma/\alpha)\varsigma_j$, so that the gap follows

$$\widehat{y}_t = \gamma\widehat{y}_{t-1} + \frac{\sigma\gamma}{\alpha}\Big[(1-\theta_d)u_t^d - \theta_\pi u_t^s - u_t^{mp}\Big]. \tag{A.39}$$

A single state thus determines every cyclical component of the system.

The cubic and the dynamics of the system. The cubic (A.38) has been obtained by matching the coefficients of a guessed solution. Before discussing determinacy, we show that

its roots are exactly the objects that govern the dynamics of the model, so that the guess entails no loss of generality. Setting the disturbances to zero, the Phillips curve (A.27) gives $E_t\widehat{\pi}_{t+1} = (\widehat{\pi}_t - \kappa\widehat{y}_t)/\beta$, and Eq. (A.30) gives $E_t\widehat{y}_{t+1} = (\widehat{y}_t - \alpha\widehat{y}_{t-1} + \sigma\theta_\pi\widehat{\pi}_t)/(1-\alpha)$. Stacking $x_t \equiv (\widehat{y}_{t-1}, \widehat{y}_t, \widehat{\pi}_t)'$, the model can be written as the first-order system

$$E_t x_{t+1} = A\, x_t, \qquad A = \begin{pmatrix} 0 & 1 & 0 \\ -\dfrac{\alpha}{1-\alpha} & \dfrac{1}{1-\alpha} & \dfrac{\sigma\theta_\pi}{1-\alpha} \\ 0 & -\dfrac{\kappa}{\beta} & \dfrac{1}{\beta} \end{pmatrix}. \tag{A.40}$$

Expanding the determinant of $A - \gamma I$ along the first row,

$$\det(A-\gamma I) = -\gamma\Big[\Big(\frac{1}{1-\alpha} - \gamma\Big)\Big(\frac{1}{\beta} - \gamma\Big) + \frac{\sigma\kappa\theta_\pi}{\beta(1-\alpha)}\Big] + \frac{\alpha}{1-\alpha}\Big(\frac{1}{\beta} - \gamma\Big),$$

and multiplying by $\beta(1-\alpha)$,

$$\beta(1-\alpha)\det(A-\gamma I) = -\gamma\Big[\big(1-(1-\alpha)\gamma\big)(1-\beta\gamma) + \sigma\kappa\theta_\pi\Big] + \alpha(1-\beta\gamma) = -F(\gamma), \tag{A.41}$$

where the last equality follows by expanding the products and collecting powers of γ. The roots of the cubic are therefore the eigenvalues of A.

Determinacy. The vector x_t contains one predetermined variable, $\widehat{y}_{t-1}$, and two non-predetermined ones, $\widehat{y}_t$ and $\widehat{\pi}_t$. The model therefore has a unique non-explosive equilibrium if and only if A has exactly one eigenvalue inside the unit circle and two outside (Blanchard and Kahn, 1980), that is, by Eq. (A.41), if and only if the cubic F has exactly one root inside the unit circle and two outside. The guess (A.31)–(A.32) evaluated at the stable root satisfies the model and is non-explosive; by uniqueness, it is the equilibrium. We now prove that F has exactly one root inside the unit circle.

Proposition A.1 *Let $\alpha \in (0,1)$, $\beta \in (0,1)$ and $\sigma, \kappa > 0$, and suppose the response to inflation is active, $\theta_\pi > 0$. Then the characteristic cubic*

$$F(\gamma) = \beta(1-\alpha)\,\gamma^3 - (1-\alpha+\beta)\,\gamma^2 + \big(1+\alpha\beta+\sigma\kappa\theta_\pi\big)\gamma - \alpha$$

has exactly one root inside the unit circle and two roots strictly outside it. Since $\widehat{y}_{t-1}$ is the only predetermined variable, the model is therefore determinate: it admits a unique non-explosive rational-expectations equilibrium, the solution (A.31)–(A.32).

Proof. The leading coefficient $\beta(1-\alpha)$ of F is strictly positive (here $\beta > 0$ and $\alpha < 1$), and over $\mathbb{C}$ the cubic has three roots $\gamma_1, \gamma_2, \gamma_3$, counted with multiplicity; dividing out the leading coefficient gives the factorisation

$$F(\gamma) = \beta(1-\alpha)\,(\gamma-\gamma_1)(\gamma-\gamma_2)(\gamma-\gamma_3). \tag{A.42}$$

Evaluating (A.42) at $\gamma = 1$ gives $F(1) = \beta(1-\alpha)(1-\gamma_1)(1-\gamma_2)(1-\gamma_3)$, while the polynomial directly gives $F(1) = \sigma\kappa\theta_\pi$. At $\gamma = 0$ the two forms must also agree: the characteristic cubic gives $F(0) = -\alpha$ and (A.42) gives $-\beta(1-\alpha)\gamma_1\gamma_2\gamma_3$, so the product of the roots is

$$P := \gamma_1\gamma_2\gamma_3 = \frac{\alpha}{\beta(1-\alpha)} > 0. \tag{A.43}$$

Since F is continuous with $F(0) = -\alpha < 0$ and $F(1) = \sigma\kappa\theta_\pi > 0$, the intermediate value theorem yields a real root of F in $(0,1)$; in particular F has at least one real root, and the roots of F are either all real, or one real together with a complex-conjugate pair.

Case 1: all three roots real. First we show that F has no root in $(-\infty, 0]$. For $\gamma \le 0$ the derivative $F'(\gamma) = 3\beta(1-\alpha)\gamma^2 - 2(1-\alpha+\beta)\gamma + (1+\alpha\beta+\sigma\kappa\theta_\pi)$ is a sum of two non-negative terms and a strictly positive constant, so $F' > 0$ on $(-\infty, 0]$. Since $F(0) = -\alpha < 0$ (as $\alpha > 0$), this gives $F(\gamma) < 0$ for every $\gamma \le 0$; hence F has no root in $(-\infty, 0]$, and all three roots are positive. Let γ^* be the smallest of them and γ_2, γ_3 the other two, so that $0 < \gamma^* \le \gamma_2, \gamma_3$ and $\gamma^* \in (0,1)$. Evaluating (A.42) at $\gamma = 1$,

$$F(1) = \beta(1-\alpha)(1-\gamma^*)(1-\gamma_2)(1-\gamma_3) = \sigma\kappa\theta_\pi > 0 \qquad (\theta_\pi > 0),$$

and since $\beta(1-\alpha) > 0$ and $1-\gamma^* > 0$, we get $(1-\gamma_2)(1-\gamma_3) > 0$. Thus $1-\gamma_2$ and $1-\gamma_3$ have the same sign: γ_2, γ_3 are both below 1 or both above 1.

By (A.43), $\gamma_2\gamma_3 = P/\gamma^*$, so they sit above 1 when

$$\gamma^* < P = \frac{\alpha}{\beta(1-\alpha)}. \tag{A.44}$$

Here γ^* being the smallest root is decisive: no root of F lies below it, so F has no zero on $(0, \gamma^*)$, and since $F(0) = -\alpha < 0$ the cubic stays negative there, i.e. $F < 0$ on $(0, \gamma^*)$. Consequently any real point at which F is positive must lie to the right of γ^*. Proving (A.44) is thereby reduced to showing that there exists a point x with $F(x) > 0$ and $x \le P$, for then $\gamma^* < x \le P$. Take $x = \alpha/(1-\alpha)$. It does not exceed P,

$$\frac{\alpha}{1-\alpha} = \beta \cdot \frac{\alpha}{\beta(1-\alpha)} = \beta P < P \qquad (\beta < 1).$$

Moreover, we show $F(x) > 0$ by splitting off the θ_π term,

$$F(\gamma) = \underbrace{\beta(1-\alpha)\gamma^3 - (1-\alpha+\beta)\gamma^2 + (1+\alpha\beta)\gamma - \alpha}_{\theta_\pi\text{-free part}} + \sigma\kappa\theta_\pi\,\gamma,$$

the θ_π-free part factors as $(\gamma-1)(\beta\gamma-1)\big[(1-\alpha)\gamma-\alpha\big]$, whose last factor vanishes at $\gamma = \alpha/(1-\alpha)$, leaving

$$F\left(\frac{\alpha}{1-\alpha}\right) = \underbrace{0}_{\theta_\pi\text{-free part}} + \sigma\kappa\theta_\pi \cdot \frac{\alpha}{1-\alpha} > 0 \qquad (\theta_\pi > 0).$$

Thus (A.44) holds, $\gamma^* < \alpha/(1-\alpha) = \beta P < P$, so $\gamma_2\gamma_3 = P/\gamma^* > 1$. The two roots therefore lie outside the unit circle, $\gamma_2, \gamma_3 > 1$, while $\gamma^* \in (0,1)$ lies inside.

Case 2: one real root and a complex-conjugate pair. Let γ^* be the single real root, which lies inside the unit circle, and let the conjugate pair be $\gamma_2, \gamma_3 = \overline{\gamma_2}$, with $\gamma_2\gamma_3 = P/\gamma^*$ by (A.43). It remains to show γ_2, γ_3 lie outside the unit circle, i.e. $|\gamma_2| > 1$. The bound (A.44) holds exactly as in Case 1, so $\gamma_2\gamma_3 = P/\gamma^* > 1$; and for a conjugate pair this product is the squared modulus,

$$|\gamma_2|^2 = \gamma_2\overline{\gamma_2} = \gamma_2\gamma_3 = P/\gamma^* > 1, \qquad \text{so} \qquad |\gamma_2| = |\gamma_3| = \sqrt{P/\gamma^*} > 1.$$

Hence γ_2, γ_3 lie strictly outside the unit circle.

In both cases $\gamma^* \in (0,1)$ is the unique root inside the unit circle and the other two lie strictly outside. As $\widehat{y}_{t-1}$ is the only predetermined variable, exactly one root inside the unit circle is the Blanchard–Kahn condition for a unique non-explosive equilibrium (Blanchard and Kahn, 1980): the minimum-state-variable solution (A.31)–(A.32), in which $\widehat{y}_{t-1}$ is the single predetermined variable. ■

Two further observations are worth recording. First, $F(0) = -\alpha < 0$ requires $\alpha > 0$; at $\alpha = 0$, $\gamma = 0$ solves Eq. (A.38), $\Phi = \kappa$, and the model degenerates to the purely forward-looking case, with the loadings still well defined in the form $b_j = \varsigma_j / D$ and $D = 1 + \sigma\theta_\pi\kappa$. Second, implicit differentiation of Eq. (A.38) at a stable root at which F crosses zero from below gives $\partial\gamma/\partial(\sigma\kappa\theta_\pi) < 0$; because Φ is increasing in γ, a stronger inflation response lowers both the persistence of the gap and the slope of the reduced-form Phillips curve.

Also notice that at $\theta_\pi = 0$ the cubic equation factors as

$$F(\gamma)\big|_{\theta_\pi=0} = (\gamma-1)(\beta\gamma-1)\big[(1-\alpha)\gamma-\alpha\big], \qquad \text{with roots } \Big\{1,\ \tfrac{1}{\beta},\ \tfrac{\alpha}{1-\alpha}\Big\}, \tag{A.45}$$

so that a root sits exactly on the unit circle and the equilibrium fails to be determinate. We therefore impose $\theta_\pi > 0$ throughout, the familiar requirement that policy respond positively to inflation; because the rule (A.18) already carries $E_t\pi_{t+1}$ with a unit coefficient, the Taylor principle here reads $\theta_\pi > 0$ rather than $\theta_\pi > 1$. Note that, with the weights normalised to sum to one, this condition involves neither the degree of backward-lookingness α nor the discount factor β.

The reduced-form Phillips curve. Equation (A.36) is a reduced-form Phillips curve. Its strength depends on the structural slope κ – as κ approaches zero the relationship becomes flat, as would arise from structural changes in pricing that sever the link between inflation and marginal costs – and on the conduct of policy.[3] Along the cycle, demand and monetary disturbances move output and inflation along the curve; supply disturbances move inflation off it and push activity in the opposite direction as the central bank leans against them, see Eq. (A.39). Episodes dominated by supply shocks therefore trace steep – and, for θ_π small,

[3]For $\theta_d = 1$ the central bank fully offsets demand shocks, suppressing the variation that traces out the curve: the mechanism in McLeay and Tenreyro (2020), in which the optimal policy response makes the reduced-form Phillips curve disappear in post-Volcker data. A stronger response to inflation operates through a complementary channel, lowering γ and hence flattening Φ, while raising the share of supply shocks in the variance of the gap.

nearly vertical or positively sloped – clouds of unemployment and inflation in the raw data, even though Φ is unchanged: this is the logic behind Figure 1 of the paper, and the reason why isolating the cyclical components matters.

Inflation expectations. Iterating Eq. (A.31), $E_t\widehat{y}_{t+h} = \gamma^h\widehat{y}_t$ for $h \geq 0$; with Eq. (A.36) and $E_t u^s_{t+h} = 0$, $E_t\widehat{\pi}_{t+h} = \Phi\gamma^h\widehat{y}_t$ for $h \geq 1$; and iterating the martingale property, $E_t\tau^\pi_{t+h} = \tau^\pi_t$. Therefore

$$E_t\pi_{t+h} = \tau^\pi_t + \Phi\gamma^h\widehat{y}_t, \qquad h \geq 1. \tag{A.46}$$

The unit loading on trend inflation is the martingale property under rational expectations, not an assumption about how expectations are measured. The cyclical loading $\Phi\gamma^h$ reflects the fact that $\widehat{y}_t$ is the only serially correlated cyclical state. Taking $h \to \infty$, $\tau^\pi_t = \lim_{h\to\infty} E_t\pi_{t+h}$: trend inflation is the long-run forecast, which is how the empirical model identifies it. Surveys measure Eq. (A.46) at $h = 4$, giving a unit loading on τ^π_t and a damped cyclical loading $\Phi\gamma^4$. There is no idiosyncratic term in Eq. (A.46) because the shocks are i.i.d., not because rational expectations are exact functions of the state: under rationality the expectation is an exact function of whatever the state is, and when the state includes persistent disturbances they appear in Eq. (A.46) too.

A.3 The trend-cycle representation and the cycle specification

The trend-cycle representation. Collecting Eqs. (A.1)–(A.9), Okun's law, the solution (A.36), and the expectations (A.46), the model implies

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ E_t\pi_{t+1} \\ i_t \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \tau^y_t \\ \tau^u_t \\ \tau^\pi_t \\ \tau^r_t \end{pmatrix} + \begin{pmatrix} 1 \\ -\delta_u(L) \\ \Phi \\ \gamma\Phi \\ \Phi(\gamma+\theta_\pi) \end{pmatrix} \widehat{y}_t + \begin{pmatrix} 0 \\ 0 \\ u^s_t \\ 0 \\ \theta_\pi u^s_t + \theta_d u^d_t + u^{mp}_t \end{pmatrix}. \tag{A.47}$$

The unit loadings on τ^π_t in the expectations and interest-rate rows follow from the martingale property of trend inflation and from the Fisher equation, and the cyclical loadings follow from Eqs. (A.36) and (A.46). The last vector collects the structural wedges, in the i.i.d. benchmark

of Appendix A.2; as shown below, persistent disturbances leave the cyclical loadings unchanged but add a further term to the inflation, expectations and interest-rate wedges. The empirical specification of Section 3 is the flexible counterpart of Eq. (A.47): it gives every observable a wedge of its own and lets each load the common cycle through a distributed lag, so that measurement error, misspecification, and the disturbances the stylised model does not name are absorbed rather than forced onto the common cycle.

The specification of the common cycle. It remains to comment on the process adopted for the common cycle. Following the trend-cycle literature in the tradition of Harvey (1985), Section 3 models the common business cycle component as an ARMA(2,1), the simplest process that displays pseudo-cyclical behaviour. This choice is a statistical device: we do not derive it from the stylised model, and nothing in the empirical specification depends on the autoregressive structure of the structural disturbances. It is nonetheless worth observing that ARMA(2,1) is exactly the order of the reduced form the model delivers once a structural disturbance is allowed to be persistent, so that the statistical choice and the theoretical benchmark agree on the order of the process, if not on its parameterisation.

The solution (A.39) makes the gap a first-order autoregression, but that is a property of the i.i.d. benchmark rather than of the economics. Take the leading case of a persistent demand disturbance, $u_t^d = \varrho u_{t-1}^d + \varepsilon_t^d$ with $\varrho \in (0,1)$, the remaining disturbances i.i.d. Writing Eq. (A.31) in lag form and applying the filter $(1-\varrho L)$, which annihilates the persistence of u_t^d,

$$(1-\varrho L)(1-\gamma L)\widehat{y}_t = b_d\varepsilon_t^d + b_s\big(u_t^s - \varrho u_{t-1}^s\big) + b_{mp}\big(u_t^{mp} - \varrho u_{t-1}^{mp}\big) \equiv w_t. \tag{A.48}$$

The process w_t is stationary, with $\mathrm{Var}(w_t) = b_d^2\sigma_d^2 + (1+\varrho^2)(b_s^2\sigma_s^2 + b_{mp}^2\sigma_{mp}^2)$, $\mathrm{Cov}(w_t, w_{t-1}) = -\varrho(b_s^2\sigma_s^2 + b_{mp}^2\sigma_{mp}^2)$, and $\mathrm{Cov}(w_t, w_{t-k}) = 0$ for $k \geq 2$. A stationary process whose autocovariances vanish beyond the first lag admits an invertible MA(1) representation, $w_t = (1+\vartheta L)e_t$, so that

$$(1-\varrho L)(1-\gamma L)\widehat{y}_t = (1+\vartheta L)e_t\,: \tag{A.49}$$

$\widehat{y}_t$ is then exactly an ARMA(2,1), with one autoregressive root inherited from its own endogenous persistence and the other from the persistence of the shock.

Persistent disturbances. The cross-equation restrictions on the common cycle are unaffected

by any of this: repeating the undetermined-coefficient matching with persistent disturbances leaves the conditions on $\widehat{y}_{t-1}$ unchanged, so γ and Φ are invariant to ϱ, which alters only the shock loadings.

We now let each structural disturbance be serially correlated,

$$u_t^j = \rho_j \, u_{t-1}^j + \varepsilon_t^j, \qquad |\rho_j| < 1, \qquad j \in \{d, s, mp\}, \tag{A.50}$$

with ε_t^j white noise, so that $E_t u_{t+1}^j = \rho_j u_t^j$. The equations to be solved are the Phillips curve and the IS equation into which the policy rule has been substituted,

$$\widehat{\pi}_t = \beta \, E_t \widehat{\pi}_{t+1} + \kappa \, \widehat{y}_t + u_t^s, \tag{A.51}$$

$$\widehat{y}_t = \alpha \, \widehat{y}_{t-1} + (1-\alpha) \, E_t \widehat{y}_{t+1} - \sigma\theta_\pi \widehat{\pi}_t + \sum_j m_j u_t^j, \tag{A.52}$$

where the exogenous terms of the second equation have been collected as $\sum_j m_j u_t^j$, with $m_d = \sigma(1-\theta_d)$, $m_s = 0$ and $m_{mp} = -\sigma$. We look for a solution of the form

$$\widehat{y}_t = \gamma \, \widehat{y}_{t-1} + \sum_j b_j u_t^j, \qquad \widehat{\pi}_t = \varphi \, \widehat{y}_{t-1} + \sum_j c_j u_t^j, \tag{A.53}$$

in which the four objects to be determined are the two coefficients γ, φ on the lagged gap and the two sets of loadings b_j, c_j on the disturbances. Leading Eq. (A.53) one period and taking expectations, using $E_t u_{t+1}^j = \rho_j u_t^j$,

$$E_t \widehat{y}_{t+1} = \gamma \, \widehat{y}_t + \sum_j \rho_j b_j u_t^j, \qquad E_t \widehat{\pi}_{t+1} = \varphi \, \widehat{y}_t + \sum_j \rho_j c_j u_t^j. \tag{A.54}$$

Serial correlation adds the sums on the right; these load the disturbances and not the lagged gap, and that is the property that drives everything below.

Substituting Eq. (A.54) into the Phillips curve (A.51),

$$\widehat{\pi}_t = (\beta\varphi + \kappa) \, \widehat{y}_t + \sum_j \beta\rho_j c_j u_t^j + u_t^s.$$

Write $\Phi \equiv \beta\varphi + \kappa$ and replace $\widehat{y}_t$ by its expression in Eq. (A.53),

$$\widehat{\pi}_t = \Phi\gamma\,\widehat{y}_{t-1} + \sum_j \Big[\Phi b_j + \beta\rho_j c_j + \mathbf{1}_{\{j=s\}}\Big] u_t^j.$$

Equating coefficients with the second equation of (A.53) gives one condition on the lagged gap and one on each disturbance,

$$\varphi = \Phi\gamma, \qquad c_j\,(1-\beta\rho_j) = \Phi b_j + \mathbf{1}_{\{j=s\}}. \tag{A.55}$$

Doing the same in the IS equation (A.52),

$$\widehat{y}_t = \big[\alpha + (1-\alpha)\gamma^2 - \sigma\theta_\pi\varphi\big]\widehat{y}_{t-1} + \sum_j \Big[(1-\alpha)(\gamma+\rho_j)b_j - \sigma\theta_\pi c_j + m_j\Big] u_t^j,$$

so that, equating coefficients with the first equation of (A.53),

$$\gamma = \alpha + (1-\alpha)\gamma^2 - \sigma\theta_\pi\varphi, \qquad b_j\big[1-(1-\alpha)(\gamma+\rho_j)\big] + \sigma\theta_\pi c_j = m_j. \tag{A.56}$$

The system splits. The two conditions on the lagged gap, the first of (A.55) and the first of (A.56), involve only γ and φ: no ρ_j, and no b_j or c_j. Eliminating $\varphi = \Phi\gamma$ between them, and using $\Phi = \beta\Phi\gamma + \kappa$, that is $\Phi = \kappa/(1-\beta\gamma)$, leaves

$$\gamma\Big[1-(1-\alpha)\gamma + \sigma\theta_\pi\Phi\Big] = \alpha, \tag{A.57}$$

which is the same equation for γ obtained under i.i.d. disturbances, and which multiplied by $(1-\beta\gamma)$ is the cubic $F(\gamma) = 0$ of Eq. (A.38). The persistence of the disturbances therefore leaves unchanged the autoregressive coefficient of the gap, the slope Φ, and with them the determinacy argument.

What persistence does change are the loadings on the disturbances. The second conditions in (A.55) and (A.56) are two linear equations in (b_j, c_j) for each j separately; solving them,

$$b_j = \frac{m_j(1-\beta\rho_j) - \sigma\theta_\pi\mathbf{1}_{\{j=s\}}}{\big[1-(1-\alpha)(\gamma+\rho_j)\big](1-\beta\rho_j) + \sigma\theta_\pi\Phi}, \qquad c_j = \frac{\Phi b_j + \mathbf{1}_{\{j=s\}}}{1-\beta\rho_j}. \tag{A.58}$$

Both depend on ρ_j, and on ρ_j only: each disturbance's loadings move with its own persistence and are unaffected by the persistence of the others.

Finally, the effect on the observables. The second condition in Eq. (A.55) gives $\Phi b_j = c_j(1-\beta\rho_j) - \mathbf{1}_{\{j=s\}}$, so that

$$c_j - \Phi b_j = \beta\rho_j c_j + \mathbf{1}_{\{j=s\}}. \tag{A.59}$$

Using the first equation of (A.53) to write $\gamma\widehat{y}_{t-1} = \widehat{y}_t - \sum_j b_j u_t^j$, substituting into the second and applying Eq. (A.59), and then substituting the result into the policy rule together with Eq. (A.54),

$$\widehat{\pi}_t = \Phi\,\widehat{y}_t + \beta\sum_j \rho_j c_j u_t^j + u_t^s, \tag{A.60}$$

$$\widehat{\imath}_t = \Phi(\gamma+\theta_\pi)\,\widehat{y}_t + (1+\beta\theta_\pi)\sum_j \rho_j c_j u_t^j + \theta_\pi u_t^s + \theta_d u_t^d + u_t^{mp}. \tag{A.61}$$

The coefficients on the contemporaneous gap are Φ and $\Phi(\gamma+\theta_\pi)$, and the remaining terms in u_t^s, u_t^d and u_t^{mp} are those of the i.i.d. case: everything that serial correlation adds is the single sum $\sum_j \rho_j c_j u_t^j$, which is the part of expected cyclical inflation not accounted for by the gap and which vanishes when the disturbances are i.i.d. In the representation of Eq. (A.47), therefore, the vector of loadings on the common cycle is unchanged and only the last vector moves.

B Adaptive Metropolis-Within-Gibbs

B.1 Algorithm

The estimation algorithm is an improved version of the Metropolis-Within-Gibbs in Hasenzagl et al. (2022) that employs the Single Component Adaptive Metropolis proposed in Haario et al. (2005).

This hybrid algorithm is structured in two blocks: (1) a Single Component Adaptive Metropolis (Haario et al., 2005) step for the estimation of the state-space parameters, (2) a Gibbs sampler (Koopman and Durbin, 2000; Jarociński, 2015) to draw the unobserved states conditional on the model parameters. Since we have non-stationary unobserved states, we use the Kalman filter with exact diffuse initial conditions (Koopman and Durbin, 2000; Durbin and Koopman, 2012) to compute the log-likelihood of the model. Finally, we use the priors of Table 1 in the main text, which follow Hasenzagl et al. (2022) with two exceptions: the initial trend levels are centred on the first observation of each series rather than on zero, and the variance of the innovation to trend output growth, a parameter absent from their model, takes a tighter Inverse-Gamma$(3, 0.005)$. For the 1946–1984 subsample, the cyclical components take a training-sample prior, which we discuss in Appendix C.1.

Algorithm: Adaptive Metropolis-Within-Gibbs

- **Initialisation**

  Let $\mathscr{K} := \{1, \ldots, n_k\}$ and denote as $\mathbf{P}(\mathscr{K})$ a function that returns a random permutation of $\mathscr{K}$ (uniformly taken from the full set of permutations of $\mathscr{K}$). Let also $\boldsymbol{\theta}_0$ be an n_k-dimensional vector corresponding to the initial value for the Metropolis parameters. This vector is associated with a high posterior mass.

- **Single Component Adaptive Metropolis**

  let $m = 1$

  for $j = 1, \ldots, 10000$

  let $\mathbf{S}_j = \mathbf{P}(\mathscr{K})$

for each k in $\mathbf{S}_j$

1. *Adaptation:* Update the standard deviation of the proposal distribution

$$\sigma_{k,j} = \begin{cases} 1 & \text{if } j \leq 10, \\ \exp\big(\alpha_{k,j-1} - 0.44\big)\sigma_{k,j-1} & \text{otherwise,} \end{cases}$$

   where $\alpha_{k,j-1}$ is the acceptance rate at iteration $j-1$, for the parameter at position $S_{k,j}$. The value 0.44 corresponds to 44%, the standard target acceptance rate for single component Metropolis algorithms.

2. *New candidate:* Generate a candidate vector of parameters $\boldsymbol{\theta}_m^*$ such that

$$\theta_{l,m}^* = \begin{cases} \theta_{l,m-1} & \text{if } l \neq k, \\ \underline{\theta} \overset{iid}{\sim} \mathcal{N}\big(\theta_{l,m-1}, \sigma_{k,j}\big) & \text{otherwise,} \end{cases}$$

   for $l = 1, \ldots, n_k$.

3. *Accept-reject:* Set

$$\boldsymbol{\theta}_m = \begin{cases} \boldsymbol{\theta}_m^* & \text{accept with probability } \eta_m, \\ \boldsymbol{\theta}_{m-1} & \text{reject with probability } 1 - \eta_m, \end{cases}$$

   where

$$\eta_m := \min\left(1, \frac{p\left[\mathbf{Y} \mid \mathbf{f}(\boldsymbol{\theta}_m^*)^{-1}\right] p\left[\mathbf{f}(\boldsymbol{\theta}_m^*)^{-1}\right] J(\boldsymbol{\theta}_m^*)}{p\left[\mathbf{Y} \mid f(\boldsymbol{\theta}_{m-1})^{-1}\right] p\left[\mathbf{f}(\boldsymbol{\theta}_{m-1})^{-1}\right] J\left[\boldsymbol{\theta}_{m-1}\right]}\right),$$

   $\mathbf{f}$ and J are defined below.

4. *Increase counter:* Increase m by one.

- **Gibbs sampling**

  For $j > 5000$ (burn-in period), apply the univariate approach for multivariate time series of Koopman and Durbin (2000) to the simulation smoother proposed in Durbin and Koopman (2002) to sample the unobserved states, conditional on the model parameters. In doing so, we follow the refinement proposed in Jarociński (2015).

- **Burn-in period**

  Discard the output of the first $j = 1, \ldots, 5000$ iterations.

- **Jacobian**

  As in Hasenzagl et al. (2022), most parameters are bounded in their support (e.g. the variance parameters must be larger than zero). In order to deal with this complexity, we transform the bounded parameters ($\boldsymbol{\Theta}$) so that the support of the transformed parameters ($\boldsymbol{\theta}$) is unbounded. Indeed, the Adaptive Metropolis-Within-Gibbs draws the model parameters in the unbounded space. At a generic iteration j, the following transformations have been applied to a generic parameter i with a Normal, Inverse-Gamma or Uniform prior:

$$\begin{aligned}
\theta_{i,j}^{N} &= \Theta_{i,j}^{N} \\
\theta_{i,j}^{IG} &= \ln(\Theta_{i,j}^{IG} - a_i) \\
\theta_{i,j}^{U} &= \ln\left(\frac{\Theta_{i,j}^{U} - a_i}{b_i - \Theta_{i,j}^{U}}\right),
\end{aligned}$$

  where a_i and b_i are the lower and the upper bounds for the i-th parameter. These transformations are functions $f(\Theta) = \theta$, with inverses $f(\theta)^{-1} = \Theta$ given by:

$$\begin{aligned}
\Theta_{i,j}^{N} &= \theta_{i,j}^{N} \\
\Theta_{i,j}^{IG} &= \exp(\theta_{i,j}^{IG}) + a_i \\
\Theta_{i,j}^{U} &= \frac{a_i + b_i \, \exp(\theta_{i,j}^{U})}{1 + \exp(\theta_{i,j}^{U})}.
\end{aligned}$$

  These transformations must be taken into account when evaluating the natural logarithm

of the prior densities by adding the Jacobians of the transformations of the variables:

$$\ln\left(\frac{d\Theta_{i,j}^{N}}{d\theta_{i,j}^{N}}\right) = 0$$

$$\ln\left(\frac{d\Theta_{i,j}^{IG}}{d\theta_{i,j}^{IG}}\right) = \theta_{i,j}^{IG}$$

$$\ln\left(\frac{d\Theta_{i,j}^{U}}{d\theta_{i,j}^{U}}\right) = \ln(b_i - a_i) + \theta_{i,j}^{U} - 2\ln(1 + \exp(\theta_{i,j}^{U})).$$

C Subsample Estimates

This appendix reports the full set of results that underlie the stability analysis of Section 5.1, obtained by re-estimating the model over the Great Inflation (1946–84) and the Great Moderation and after (1985–2019). For each subsample we show the same figures as in Section 4 for the full sample. Appendix C.5 reports the corresponding charts for the pandemic period; these are not a re-estimation, but the model estimated over 1900–2019 and brought to the COVID sample.

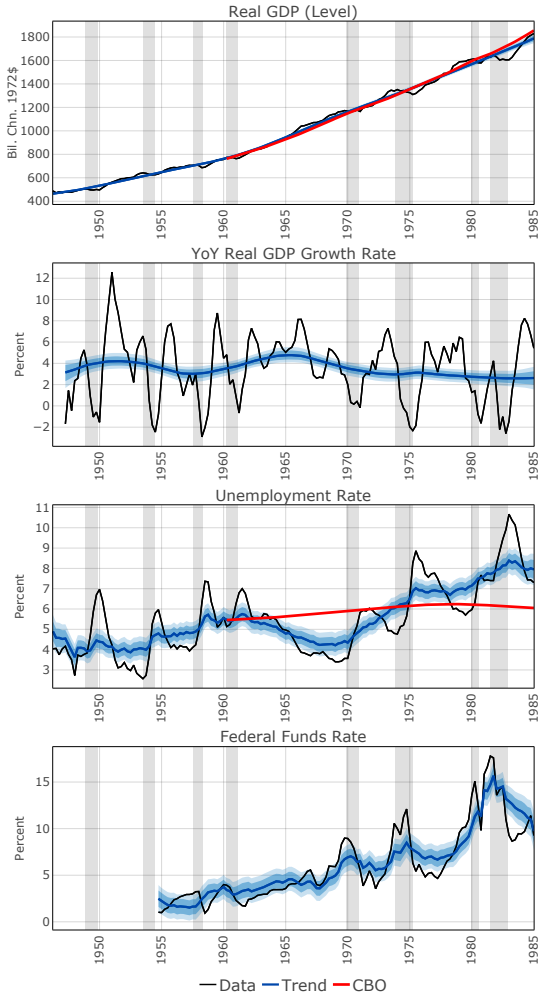
C.1 The Great Inflation: 1946–1984

The model estimated over 1946–84 departs from the baseline in two respects, motivated in Section 3.2. First, it excludes the SPF (Livingston) and Michigan surveys, together with the parameters that exist only because of them: their loadings on the common business cycle component, their idiosyncratic cycles and their survey biases. Five series remain: GDP, unemployment, headline and core CPI inflation, and the policy rate.

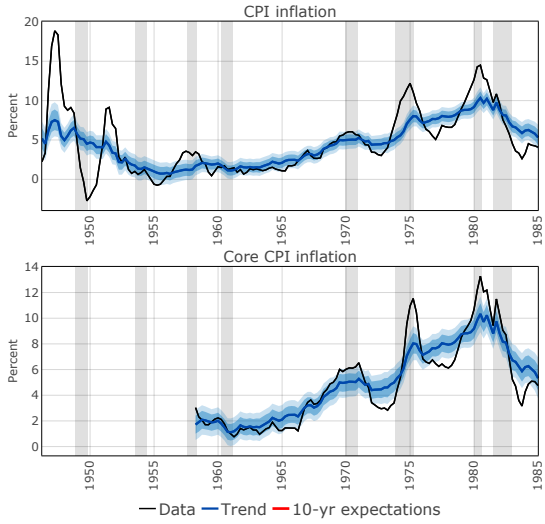
Second, the cyclical components take a training-sample prior. The training model is the baseline model restricted to GDP, unemployment and CPI inflation, estimated over 1900–1945 under the priors of Table 1. Its posterior provides the prior for three sets of parameters of the 1946–84 model: the shock variance, frequency λ and damping ρ of the common business cycle component; the loadings of unemployment and CPI inflation on the common business cycle component and on its auxiliary state; and the shock variance, frequency and damping of the idiosyncratic cycles of GDP, unemployment and CPI inflation. Each prior is centred on the training-sample posterior mean, with standard deviation equal to the training-sample posterior standard deviation: a Normal for the loadings, an Inverse-Gamma with the same mean and standard deviation for the shock variances, and a Normal truncated to the admissible region for λ and ρ. All other parameters keep the priors of Table 1: the trend shocks, including that of trend inflation, the initial conditions, and the loadings and idiosyncratic cycles of core inflation and of the policy rate. The training sample ends where the estimation sample begins, so the prior uses no information from 1946–84.

Figure C.1: Trend components, 1946–1984.

Panel A: Activity and Policy Rate

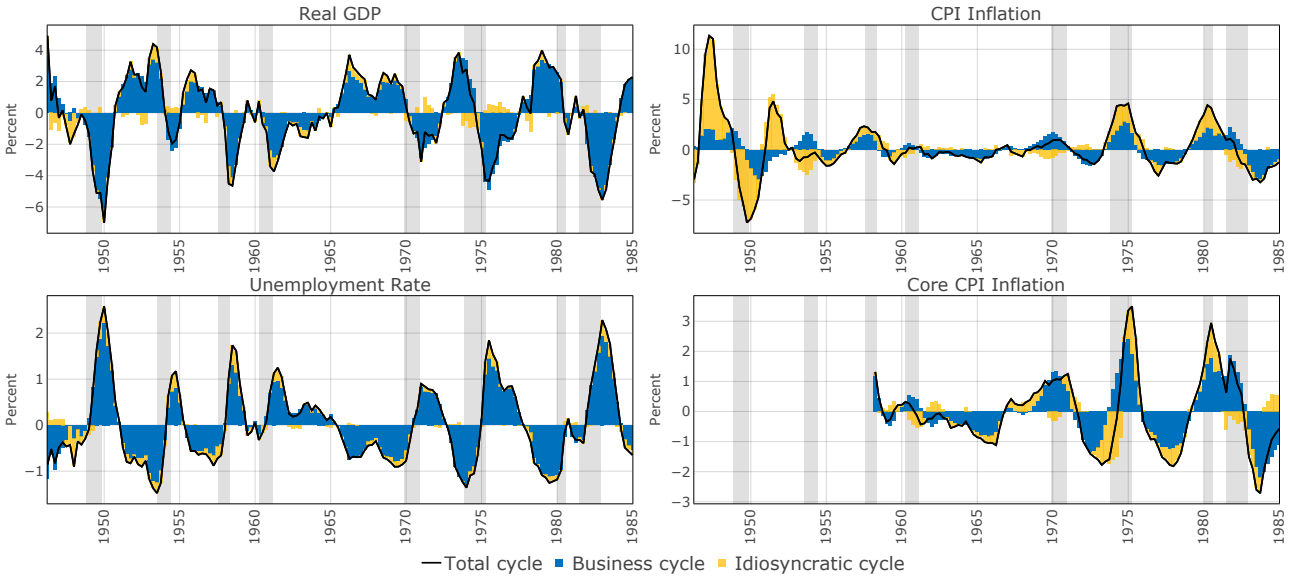


Panel B: Inflation



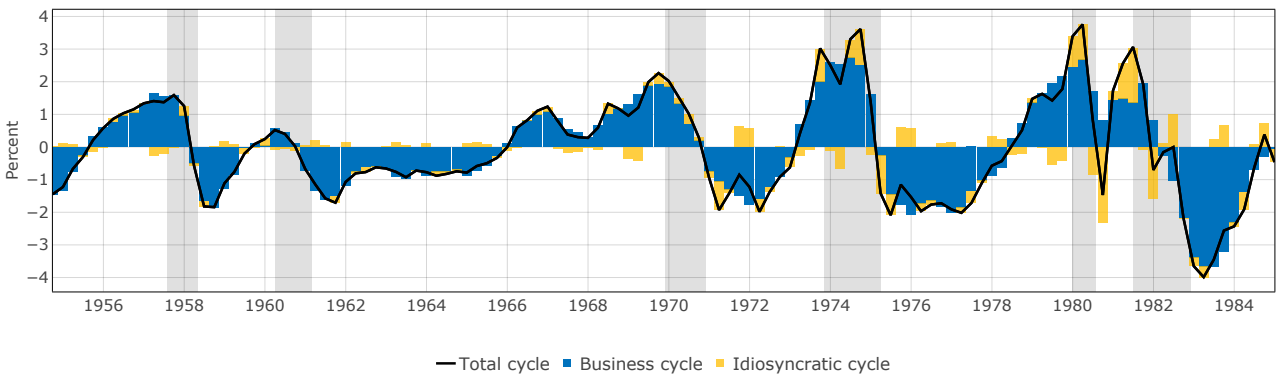
Notes: Estimated trends (in blue), with 68% (dark shade) and 90% (light shade) coverage intervals. The observed series are in black, and external comparison measures in red. The model is estimated over the sample 1946–1984.

Figure C.2: Stationary components, 1946–1984.



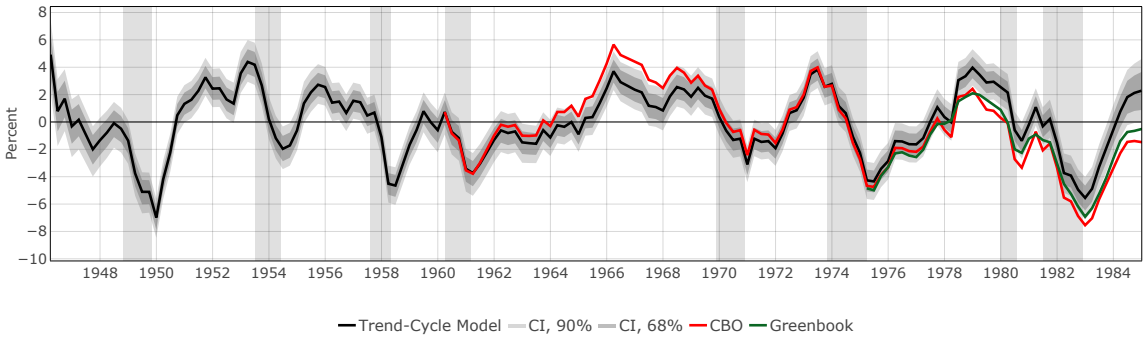
Notes: Cyclical decomposition into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line). The model is estimated over the sample 1946–1984.

Figure C.3: Cyclical components of the federal funds rate, 1946–1984.



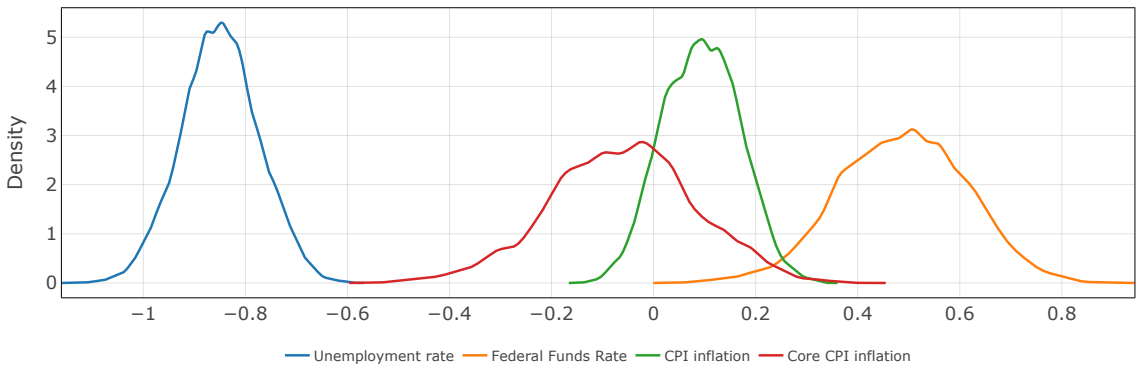
Notes: Cyclical decomposition of the federal funds rate into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line). Shaded areas are NBER recessions. The model is estimated over the sample 1946–1984.

Figure C.4: Output gap, 1946–1984.



Notes: Output gap as a percentage of potential output, with 68/90% posterior bands, against the CBO and Federal Reserve (Greenbook) estimates. Shaded areas are NBER recessions. The model is estimated over the sample 1946–1984.

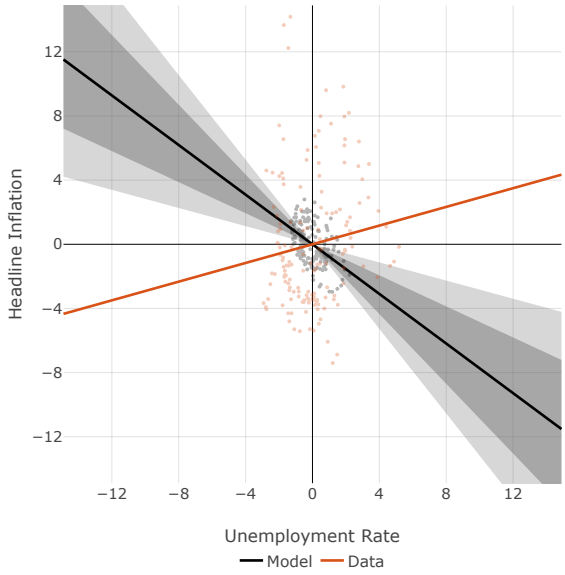
Figure C.5: Loadings on the common business cycle component, 1946–1984.



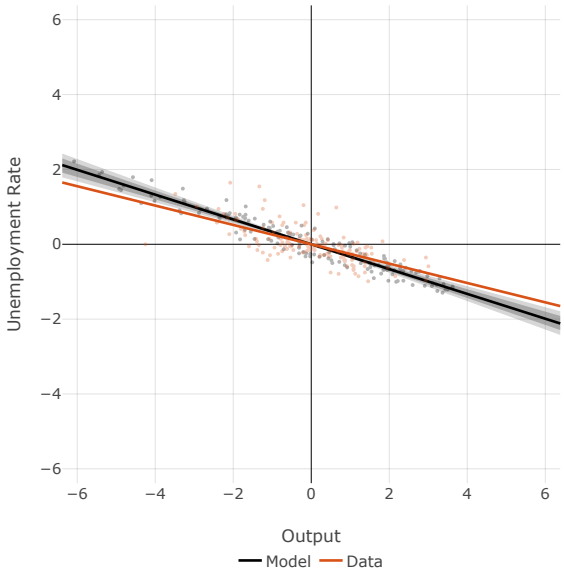
Notes: Posterior distributions of each series' loading on the common business cycle. The model is estimated over the sample 1946–1984.

Figure C.6: Phillips curve and Okun's law, 1946–1984.

Panel A: Phillips curve



Panel B: Okun's law



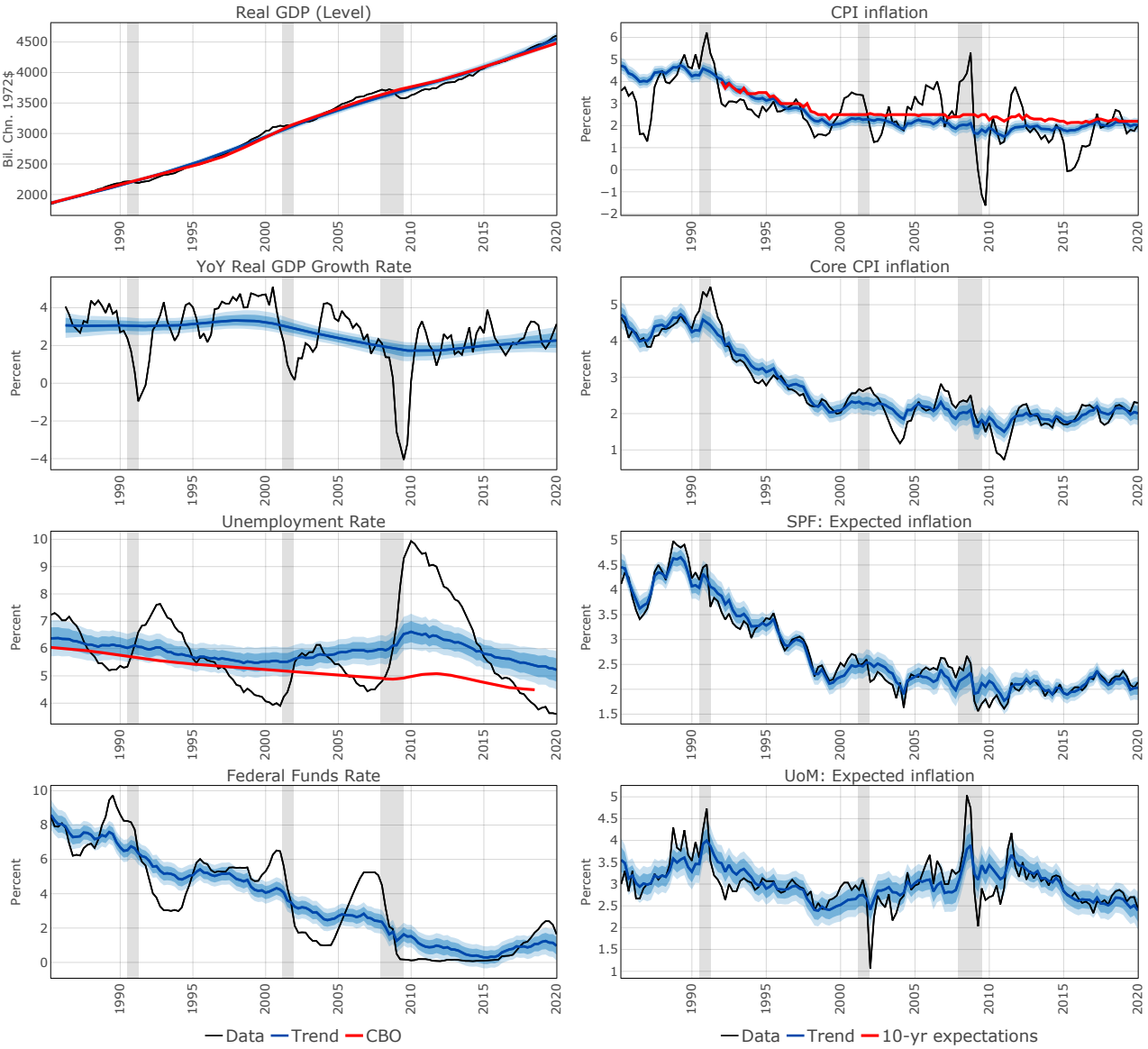
Notes: In black: the business cycle component of one variable against the business cycle component of another, their bivariate linear-regression line, and the 68/90% posterior credible band. In red: the corresponding demeaned raw data and their bivariate linear-regression line. The model is estimated over the sample 1946–1984.

C.2 The Great Moderation: 1985–2019

Figure C.7: Trend components, 1985–2019.

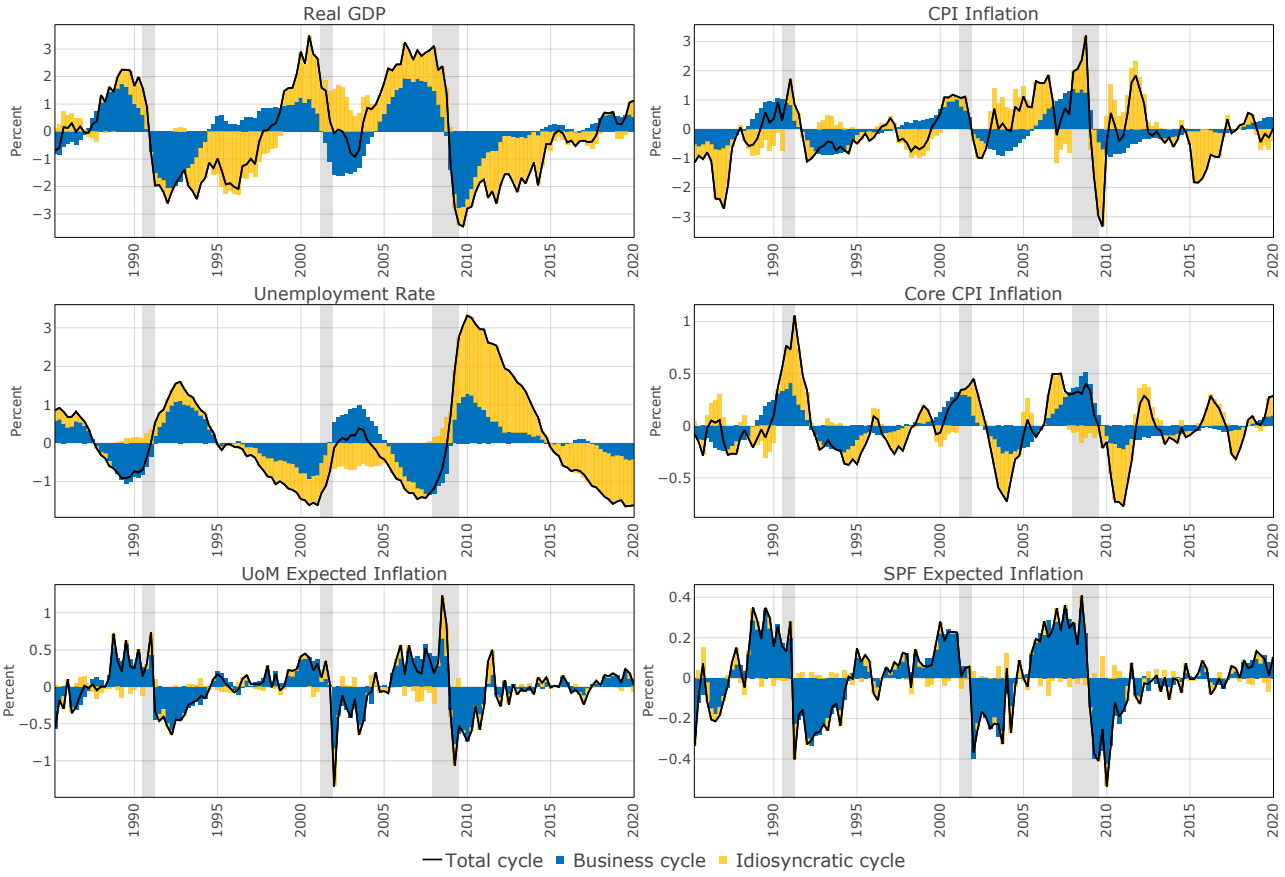
Panel A: Activity and Policy Rate

Panel B: Inflation



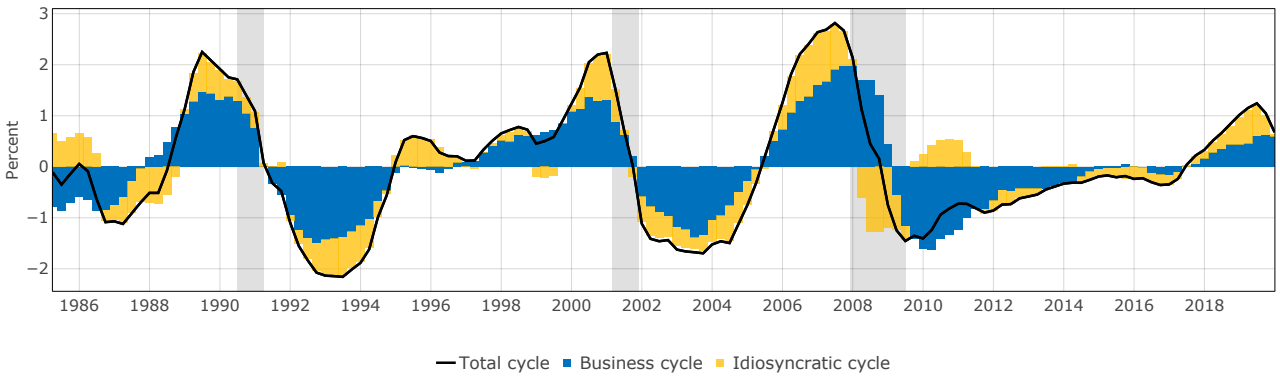
Notes: Estimated trends (in blue), with 68% (dark shade) and 90% (light shade) coverage intervals. The observed series are in black, and external comparison measures in red. The model is estimated over the sample 1985–2019.

Figure C.8: Stationary components, 1985–2019.



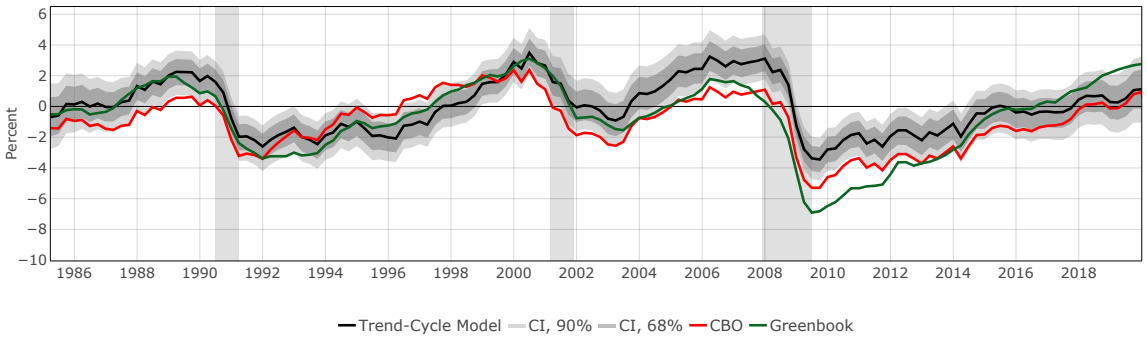
Notes: Cyclical decomposition into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line). The model is estimated over the sample 1985–2019.

Figure C.9: Cyclical components of the federal funds rate, 1985–2019.



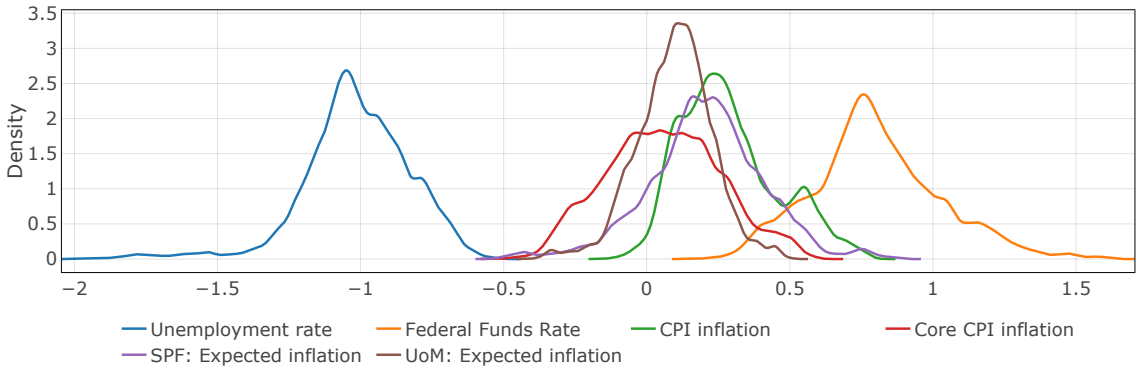
Notes: Cyclical decomposition of the federal funds rate into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line). Shaded areas are NBER recessions. The model is estimated over the sample 1985–2019.

Figure C.10: Output gap, 1985–2019.



Notes: Output gap as a percentage of potential output, with 68/90% posterior bands, against the CBO and Federal Reserve (Greenbook) estimates. Shaded areas are NBER recessions. The model is estimated over the sample 1985–2019.

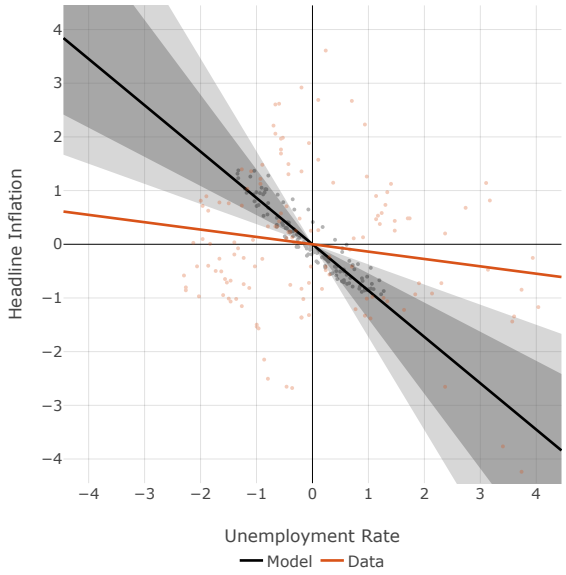
Figure C.11: Loadings on the common business cycle component, 1985–2019.



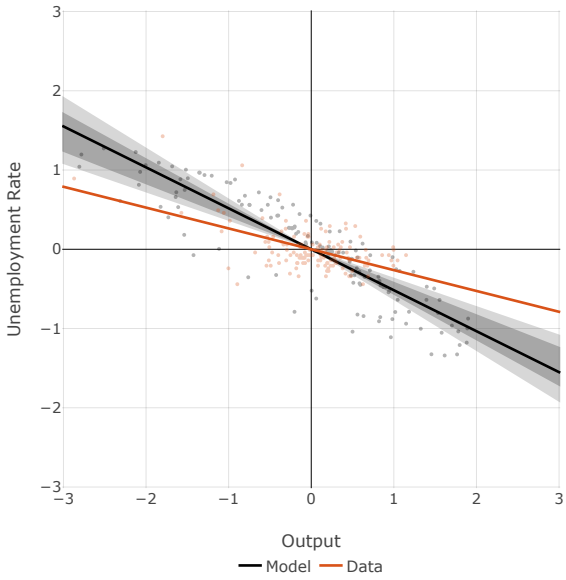
Notes: Posterior distributions of each series' loading on the common business cycle. The model is estimated over the sample 1985–2019.

Figure C.12: Phillips curve and Okun's law, 1985–2019.

Panel A: Phillips curve



Panel B: Okun's law

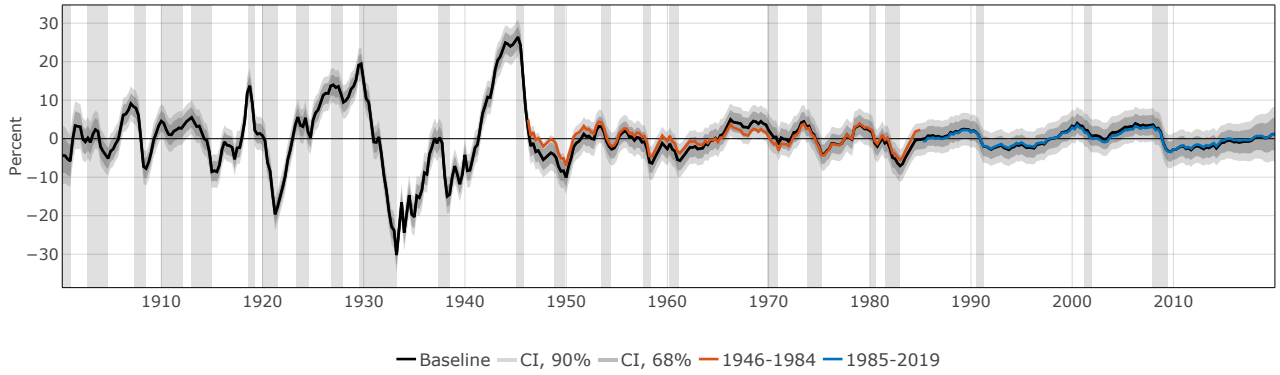


Notes: In black: the business cycle component of one variable against the business cycle component of another, their bivariate linear-regression line, and the 68/90% posterior credible band. In red: the corresponding demeaned raw data and their bivariate linear-regression line. The model is estimated over the sample 1985–2019.

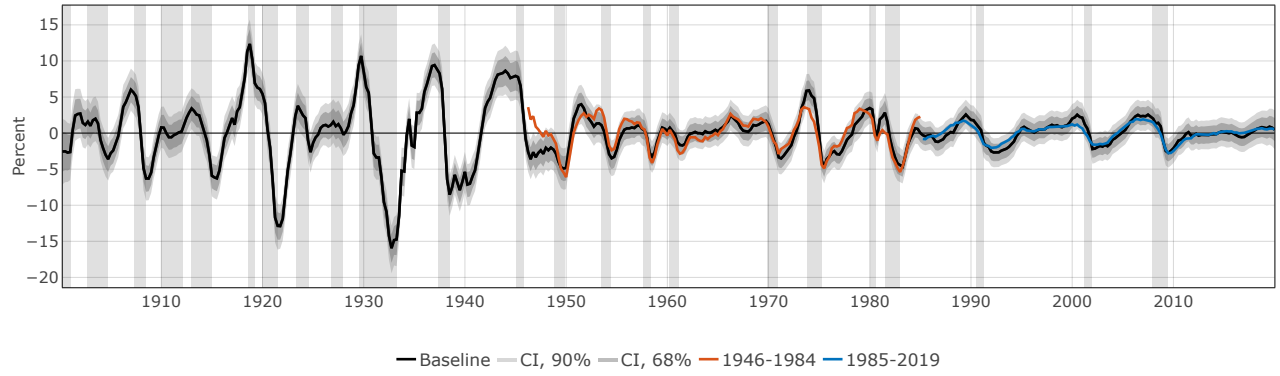
C.3 Output gap – sample comparison

Figure C.13: Output gap & common business cycle in subsamples.

Panel A: Output gap

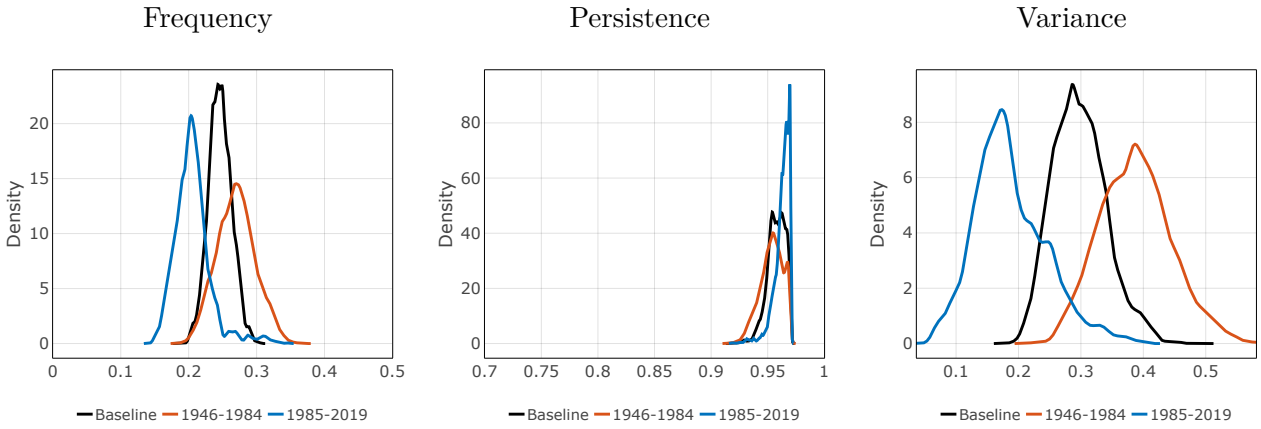


Panel B: Common business cycle



Notes: In both panels the full-sample model is in black, with its 68/90% posterior credible band in grey, against the two postwar subperiods, 1946–1984 and 1985–2019. Shaded areas are NBER recessions.

Figure C.14: Common business cycle parameters.



Notes: Posterior distributions of the common business cycle parameters across samples: the frequency λ, the persistence ρ, and the innovation variance σ^2.

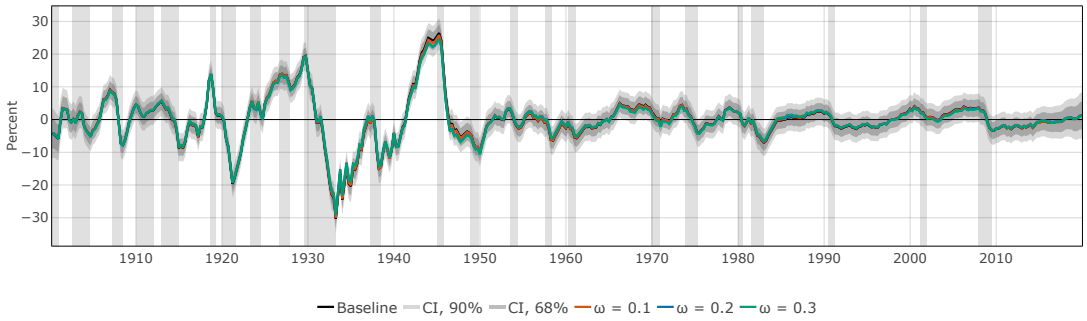
C.4 Output gap – correlated trend-cycle shocks

Table C.1: Prior and posterior of the cycle–trend correlations.

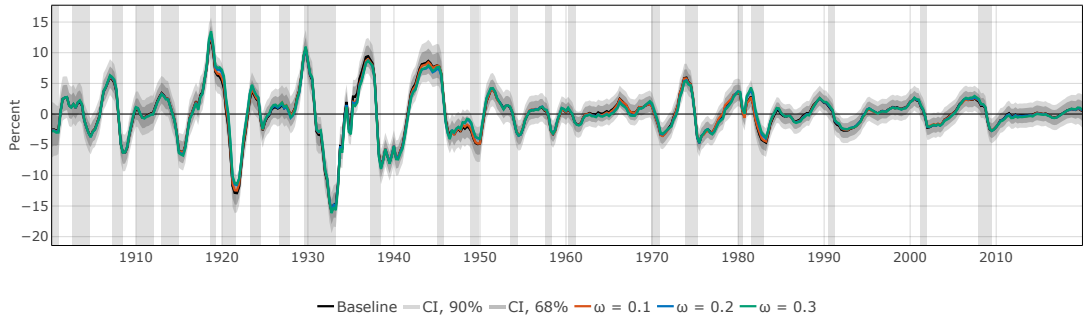
	Prior			Posterior		
ω	ρ_π	ρ_g	ρ_u	ρ_π	ρ_g	ρ_u
0.10	0.00	0.00	0.00	-0.18	-0.07	0.10
	[-0.10, 0.10]	[-0.10, 0.10]	[-0.10, 0.10]	[-0.27, -0.09]	[-0.15, 0.02]	[0.01, 0.20]
0.20	0.00	0.00	0.00	-0.52	-0.12	0.21
	[-0.19, 0.19]	[-0.19, 0.19]	[-0.19, 0.19]	[-0.62, -0.41]	[-0.23, -0.01]	[0.08, 0.34]
0.30	0.00	0.00	0.00	-0.62	-0.16	0.27
	[-0.27, 0.27]	[-0.27, 0.27]	[-0.27, 0.27]	[-0.71, -0.54]	[-0.28, -0.04]	[0.13, 0.41]

Notes: Prior and posterior of the correlations between the common business-cycle shock and the shocks to trend inflation (ρ_π), trend output growth (ρ_g) and the NAIRU (ρ_u), from the correlated trend-cycle specification estimated at each prior scale ω. Each cell reports the mean with the 68% interval beneath; the prior mean is zero by construction. The induced prior on the correlation depends on ω alone and is common to the three correlations. A posterior displaced from the prior indicates that the data are informative about the correlation.

Figure C.15: Output gap and common business cycle components with correlation between trends and cycle.



Output gap



Common business cycle

Notes: In both panels the baseline model is in black, with its 68/90% posterior credible band in grey, against the correlated trend-cycle specifications. Shaded areas are NBER recessions.

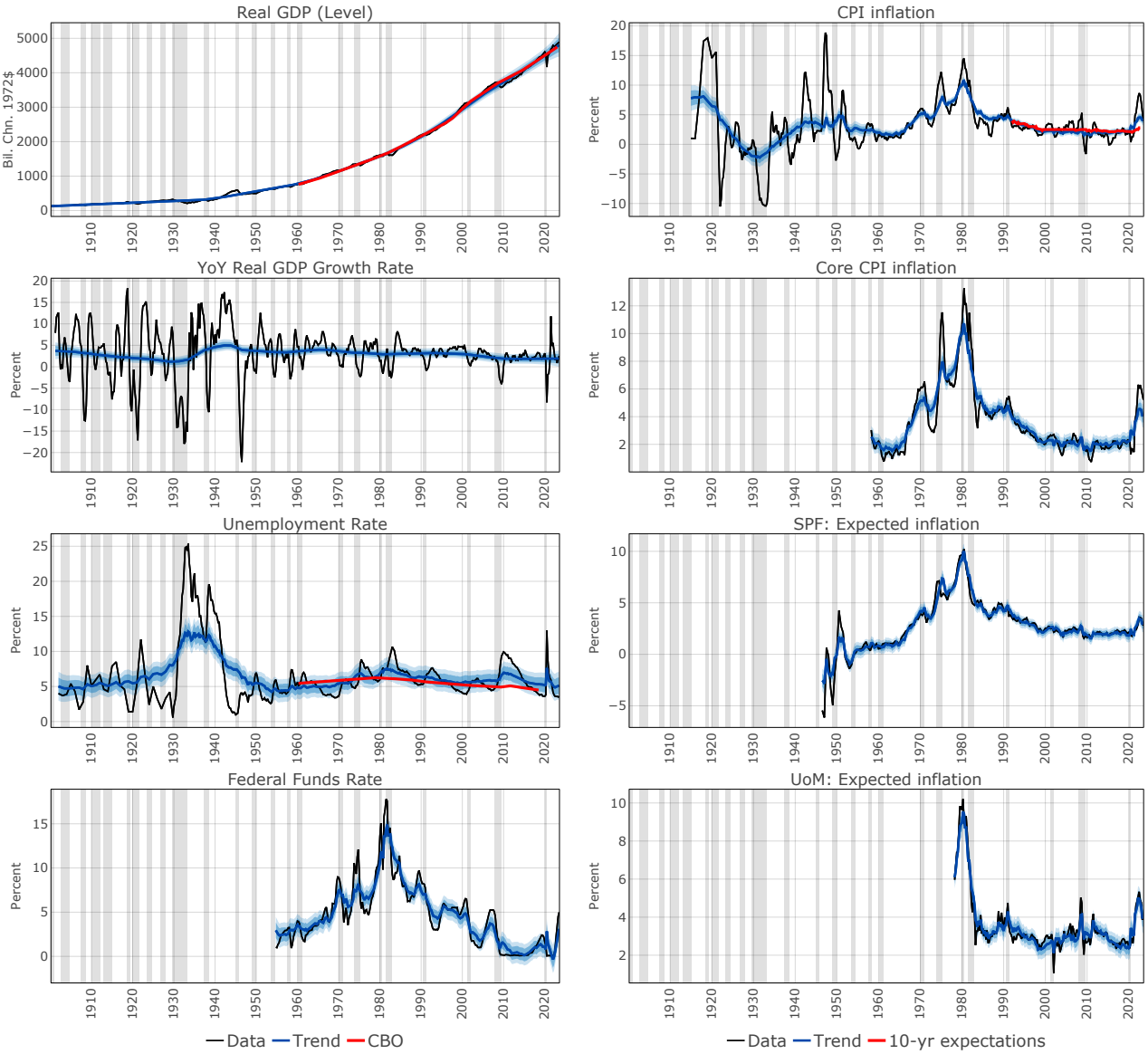
C.5 The Pandemic Period

The figures in this subsection are not obtained from a separate estimation. The model is estimated over 1900–2019, exactly as in the baseline, and is then brought to the pandemic period: the parameters are held at their full-sample posterior and the Kalman filter and smoother are run over the extended sample. What the charts show is therefore how the trends and cycles identified on a century of pre-pandemic data extrapolate through COVID.

Figure C.16: Trend components, pandemic period.

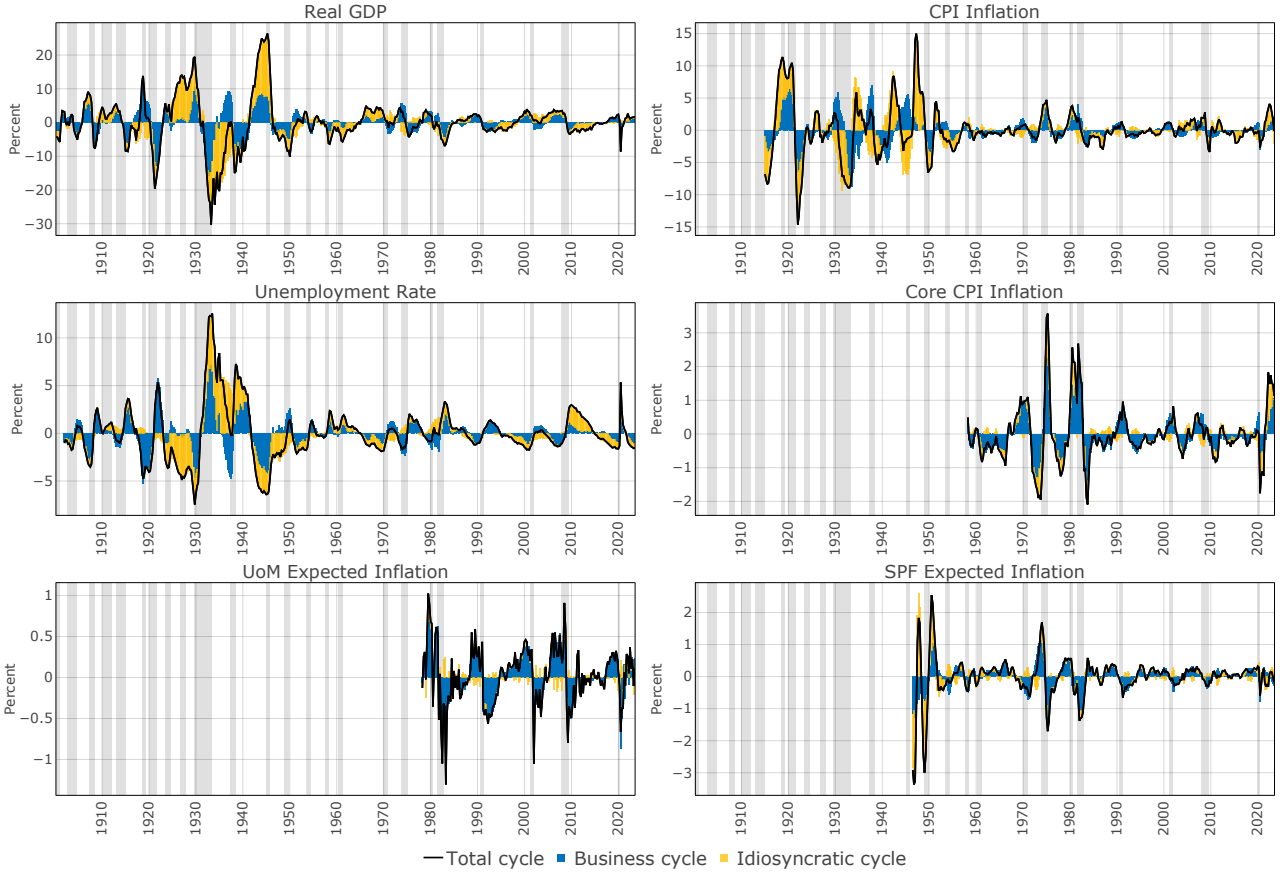
Panel A: Activity and Policy Rate

Panel B: Inflation



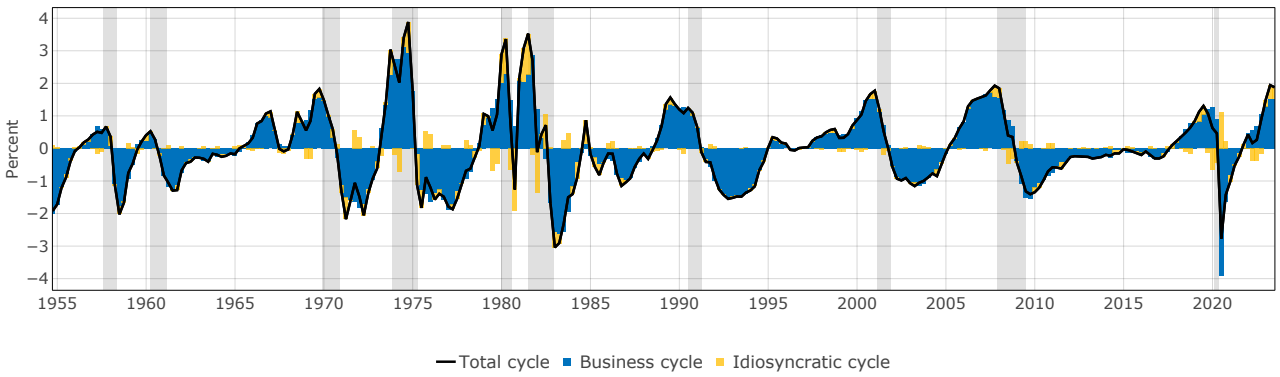
Notes: Estimated trends (in blue), with 68% (dark shade) and 90% (light shade) coverage intervals. The observed series are in black, and external comparison measures in red. The model is estimated over the sample 1900–2019 and brought to the pandemic period.

Figure C.17: Stationary components, pandemic period.



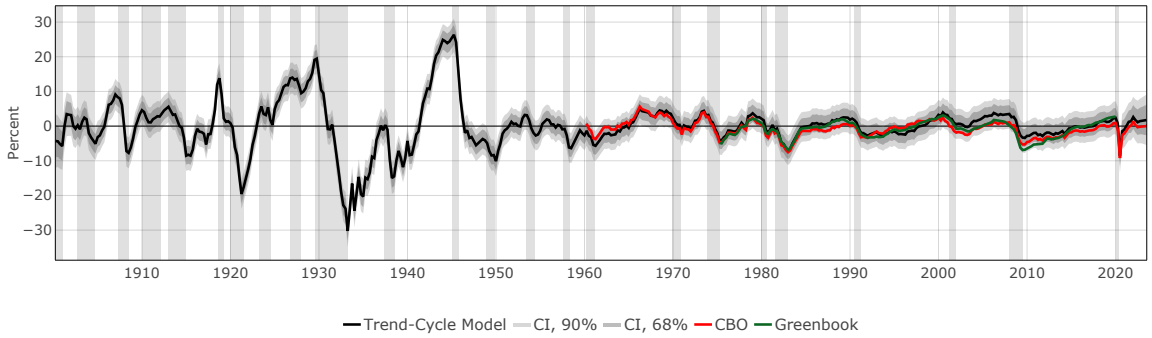
Notes: Cyclical decomposition into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line). The model is estimated over the sample 1900–2019 and brought to the pandemic period.

Figure C.18: Cyclical components of the federal funds rate, pandemic period.



Notes: Cyclical decomposition of the federal funds rate into the common business cycle (blue), the idiosyncratic component (yellow), and the total cycle (black line). Shaded areas are NBER recessions. The model is estimated over the sample 1900–2019 and brought to the pandemic period.

Figure C.19: Output gap, pandemic period.



Notes: Output gap as a percentage of potential output, with 68/90% posterior bands, against the CBO and Federal Reserve (Greenbook) estimates. Shaded areas are NBER recessions. The model is estimated over the sample 1900–2019 and brought to the pandemic period.

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